

GATE 2009, ECE Question Number 36

Abstract

This project implements and verifies a logic equation derived from a GATE 2009 ECE question using AVR-GCC and physical hardware (ATmega328P). Logical analysis determines the value of variable Z , and the expression is validated using an LED output.

1. Question and Simplification

- Given: $X = 1$
- Expression: $(X+Z)(Y+(Z+Y))(X+Z(X+Y)) = 1$
- Simplifies to: $(1)(Y+Z)(1+Z(1+Y)) = 1$
- Solving: $Z(1+Y) = 0 \Rightarrow Z = 0$

Correct Answer: (D) $Z = 0$

2. Components

Component	Qty
ATmega328P / AVR Board	1
Push Buttons (X, Y, Z)	3
LED for Output	1
Resistors (220 Ω , 10k Ω)	4
Breadboard	1
Jumper Wires	10
Laptop with AVR-GCC	1

Table 1: Components used

3. Setup and Connections

- Connect buttons to PD0 (X), PD1 (Y), PD2 (Z).
- Connect LED to PB0 with 220 Ω resistor.
- Use 10k Ω pull-down resistors for each button.
- Ensure common GND and VCC rails.

4. Logic Summary

- Inputs: X, Y, Z (via push buttons)
- Evaluated Expression: $(X+Z)(Y+Z)(1+Z(1+Y))$
- Output (LED) glows only if the expression evaluates to 1

5. Pin Mapping

- **X** – PD0 (Input)
- **Y** – PD1 (Input)
- **Z** – PD2 (Input)
- **LED** – PB0 (Output)

6. Verification Procedure

1. Set up the circuit on a breadboard using the components listed.

2. Ensure pull-down resistors are connected properly for X, Y, and Z inputs.
3. Burn the compiled AVR-GCC C code into the ATmega328P microcontroller.
4. Toggle the buttons to test various input combinations.
5. Observe LED state for each combination.
6. Verify output matches expected logic from truth table.

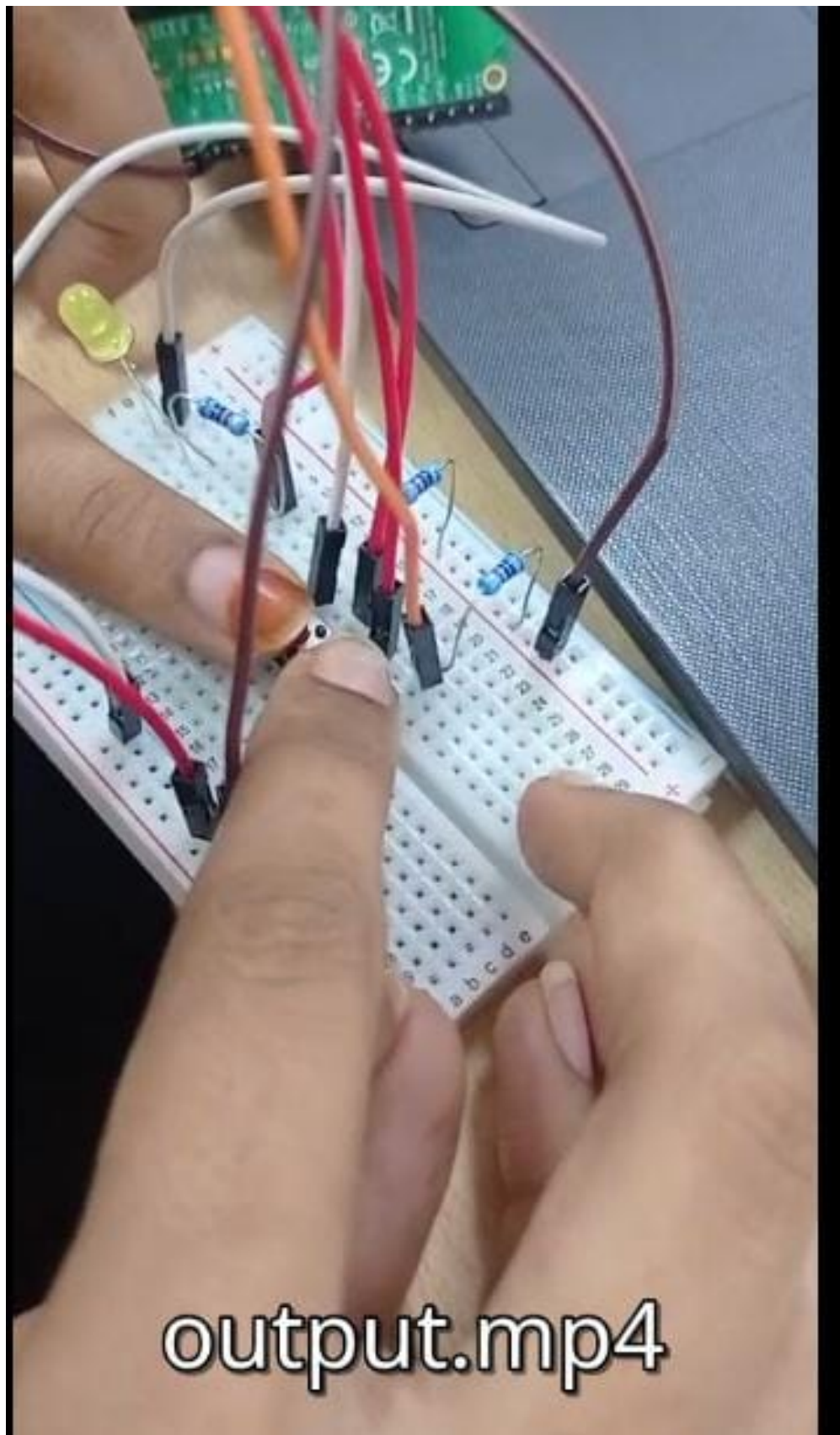
7. Analysis and Results

7.1 Sample Truth Table

X	Y	Z	Output
1	0	0	1
1	1	0	1
1	0	1	0
1	1	1	0

Table 2: Logic validation for Z

7.2 Circuit Implementation



8. Conclusion

This project successfully demonstrated the evaluation of a logical expression from GATE 2009 using AVR hardware and C programming. The experimental LED output confirmed that the function outputs 1 only when $Z = 0$, validating the theoretical simplification.

9. References

- GitHub Repository: https://github.com/Rithrishareddy/RITHRISHA_FWC/tree/main/Hardware
- GATE 2009 ECE Question Paper
- AVR-GCC Documentation and Datasheets